Definition 5.2.2. $z \in \mathbb{C}$ is an **algebraic number**, (respectively, an **algebraic integer**), if z is an eigenvalue of a matrix with rational (respectively, integer) entries.

Proposition 5.2.3. *Definitions (5.2.1) and (5.2.2) are equivalent.*

Proof. To show that the condition of Definition 5.2.2 implies the condition of Definition 5.2.1, notice that z is a root of the characteristic polynomial of the matrix (a monic polynomial with rational, respectively integer, coefficients). To establish the converse, suppose z is a root of

$$p(x) = x^n + a_1 x^{n-1} + \ldots + a_{n-1}x + a_n.$$

Then the characteristic polynomial of the following matrix (called the **companion matrix**) is $p(x)$:

$$\begin{pmatrix} 0 & 0 & 0 & \ldots & 0 & -a_n \\ 1 & 0 & 0 & \ldots & 0 & -a_{n-1} \\ 0 & 1 & 0 & \ldots & 0 & -a_{n-2} \\ & & & \vdots & & \\ 0 & 0 & 0 & \ldots & 1 & -a_1 \end{pmatrix}.$$

Since z is a root of the characteristic polynomial of this matrix, it is an eigenvalue of this matrix. □

The set of algebraic numbers is denoted by $\overline{\mathbb{Q}}$, and the set of algebraic integers is denoted by $\overline{\mathbb{Z}}$.

Proposition 5.2.4. *(i) $\overline{\mathbb{Z}}$ is a ring.*

(ii) $\overline{\mathbb{Q}}$ is a field. Namely, it is an algebraic closure of the field of rational numbers.

Proof. We will be using Definition 5.2.2. Let α be an eigenvalue of

$$\mathcal{A} \in \mathrm{Mat}_n(\mathbb{C})$$

with eigenvector v, and let β be an eigenvalue of

$$\mathcal{B} \in \mathrm{Mat}_m(\mathbb{C})$$